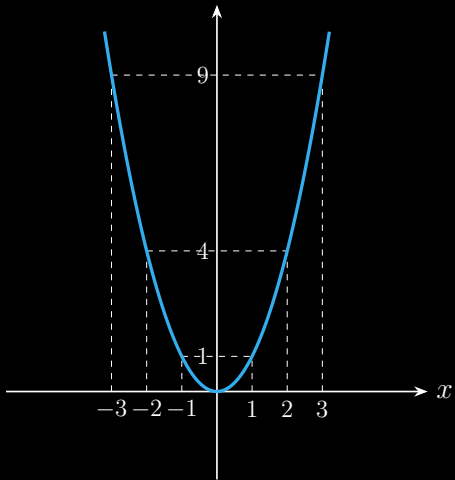
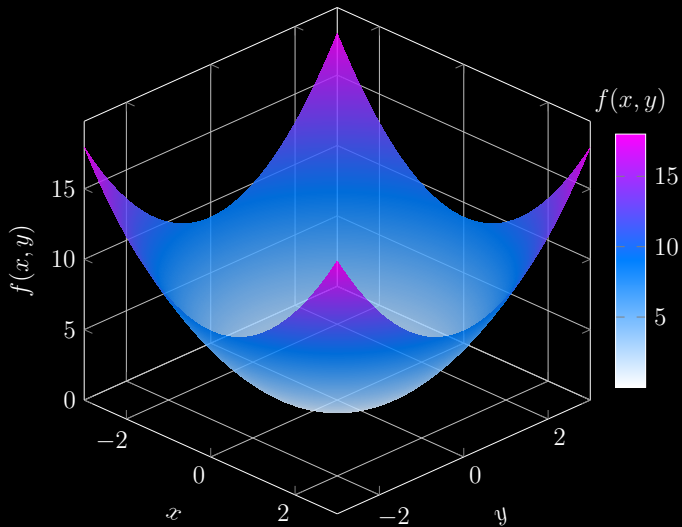


Quadratic Function

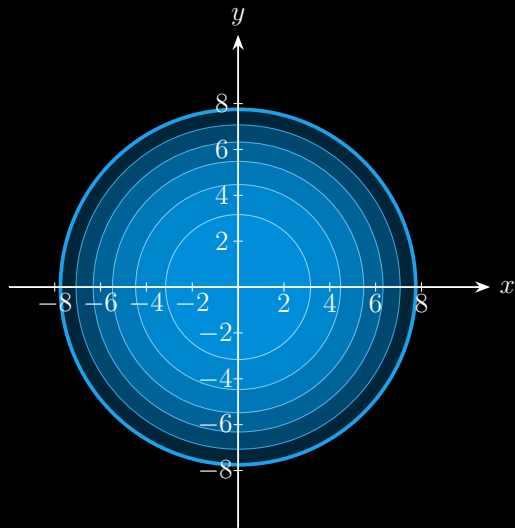
$$f(x) = x^2$$



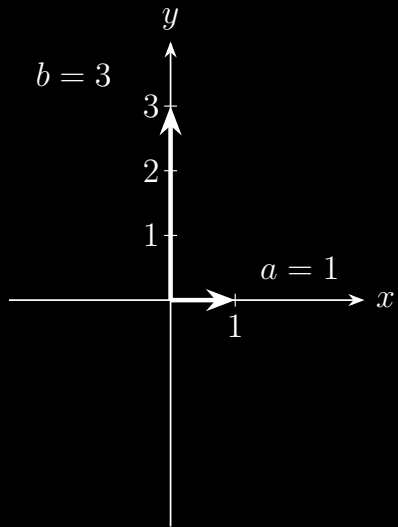
Quadratic Function $f(x, y) = x^2 + y^2$

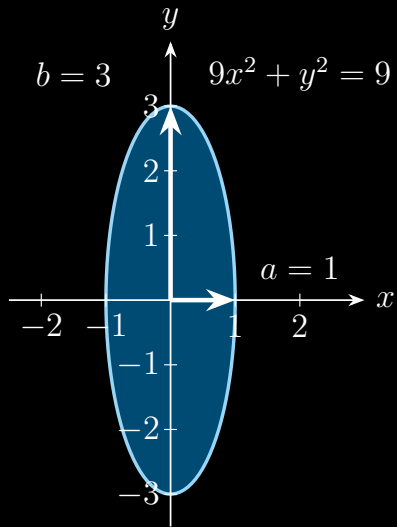


Quadratic Function $f(x, y) = x^2 + y^2$

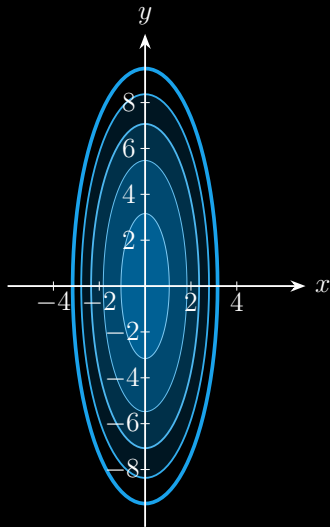


Ellipse: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

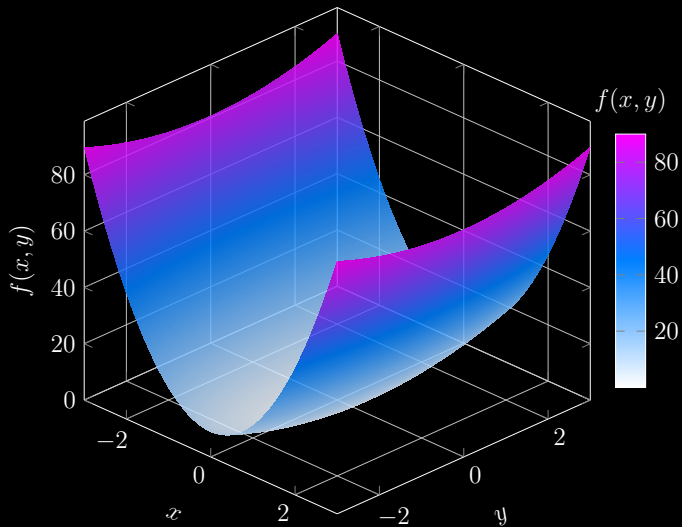




Quadratic Function $f(x, y) = 9x^2 + y^2$



Quadratic Function $f(x, y) = 9x^2 + y^2$



Quadratic Function

$$f(x, y) = 9x^2 + 4xy + y^2 = [x \ y] \underbrace{\begin{bmatrix} 9 & 2 \\ 2 & 1 \end{bmatrix}}_{\mathbf{A}} \begin{bmatrix} x \\ y \end{bmatrix}$$

Eigenvalue decomposition:

$$\mathbf{A} = \lambda_1 \mathbf{u}_1 \mathbf{u}_1^\top + \lambda_2 \mathbf{u}_2 \mathbf{u}_2^\top$$

with eigenvalues:

$$\lambda_1 = 9.47 \quad \lambda_2 = 0.53$$

and eigenvectors:

$$\mathbf{u}_1 = \begin{bmatrix} 0.97 \\ 0.23 \end{bmatrix} \quad \mathbf{u}_2 = \begin{bmatrix} -0.23 \\ 0.97 \end{bmatrix}$$

Quadratic Function

$$f(x, y) = 9x^2 + 4xy + y^2 = [x \ y] \underbrace{\begin{bmatrix} 9 & 2 \\ 2 & 1 \end{bmatrix}}_A \begin{bmatrix} x \\ y \end{bmatrix}$$

Eigenvalue decomposition:

$$A = \lambda_1 \mathbf{u}_1 \mathbf{u}_1^\top + \lambda_2 \mathbf{u}_2 \mathbf{u}_2^\top$$

with eigenvalues:

$$\lambda_1 = 9.47 \quad \lambda_2 = 0.53$$

and eigenvectors:

$$\mathbf{u}_1 = \begin{bmatrix} 0.97 \\ 0.23 \end{bmatrix} \quad \mathbf{u}_2 = \begin{bmatrix} -0.23 \\ 0.97 \end{bmatrix}$$

```
1 import numpy as np
2
3 A = np.array([[9, 2], [2, 1]])
4 eig_val, eig_vec = np.linalg.eig(A)
5
6 print('lambda_1 = ', eig_val[0])
7 print('lambda_2 = ', eig_val[1])
8 print('u_1 = ', eig_vec[:, 0])
9 print('u_2 = ', eig_vec[:, 1])
```

Quadratic Function

$$f(x, y) = 9x^2 + 4xy + y^2 = \underbrace{\begin{bmatrix} x & y \end{bmatrix}}_{\mathbf{w}^\top} \underbrace{\begin{bmatrix} 9 & 2 \\ 2 & 1 \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} x \\ y \end{bmatrix}}_{\mathbf{w}}$$

Using the eigenvalue decomposition:

$$\begin{aligned} \mathbf{w}^\top \mathbf{A} \mathbf{w} &= \mathbf{w}^\top (\lambda_1 \mathbf{u}_1 \mathbf{u}_1^\top + \lambda_2 \mathbf{u}_2 \mathbf{u}_2^\top) \mathbf{w} \\ &= \lambda_1 (\mathbf{u}_1^\top \mathbf{w})^2 + \lambda_2 (\mathbf{u}_2^\top \mathbf{w})^2 \end{aligned}$$

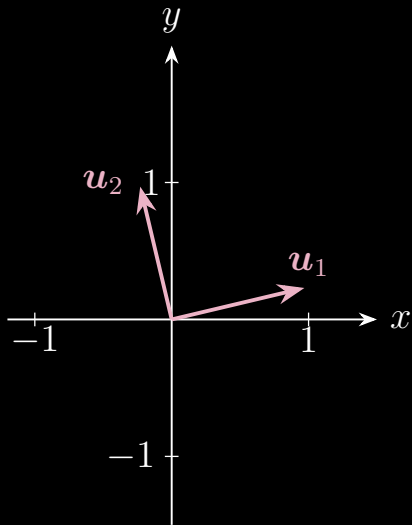
with eigenvalues and eigenvectors:

$$\lambda_1 = 9.47 \quad \mathbf{u}_1 = \begin{bmatrix} 0.97 \\ 0.23 \end{bmatrix} \quad \lambda_2 = 0.53 \quad \mathbf{u}_2 = \begin{bmatrix} -0.23 \\ 0.97 \end{bmatrix}$$

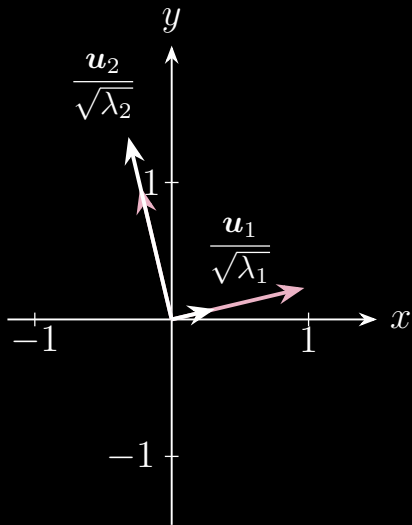
If $f(x, y) = 1$, then the ellipse is

$$\lambda_1(\mathbf{u}_1^\top \mathbf{w})^2 + \lambda_2(\mathbf{u}_2^\top \mathbf{w})^2 = 1 \quad \Rightarrow \quad \frac{(\mathbf{u}_1^\top \mathbf{w})^2}{(1/\sqrt{\lambda_1})^2} + \frac{(\mathbf{u}_2^\top \mathbf{w})^2}{(1/\sqrt{\lambda_2})^2} = 1$$

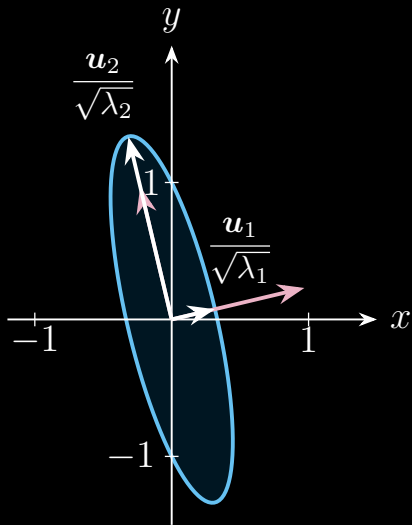
Ellipse $\frac{(\mathbf{u}_1^\top \mathbf{w})^2}{(1/\sqrt{\lambda_1})^2} + \frac{(\mathbf{u}_2^\top \mathbf{w})^2}{(1/\sqrt{\lambda_2})^2} = 1$



Ellipse $\frac{(\mathbf{u}_1^\top \mathbf{w})^2}{(1/\sqrt{\lambda_1})^2} + \frac{(\mathbf{u}_2^\top \mathbf{w})^2}{(1/\sqrt{\lambda_2})^2} = 1$



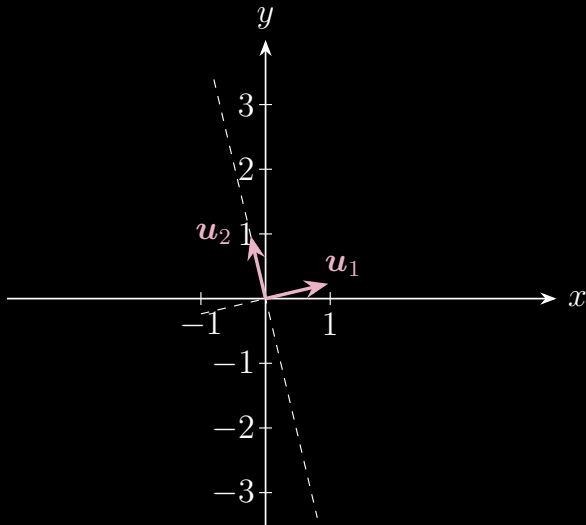
Ellipse $\frac{(\mathbf{u}_1^\top \mathbf{w})^2}{(1/\sqrt{\lambda_1})^2} + \frac{(\mathbf{u}_2^\top \mathbf{w})^2}{(1/\sqrt{\lambda_2})^2} = 1$



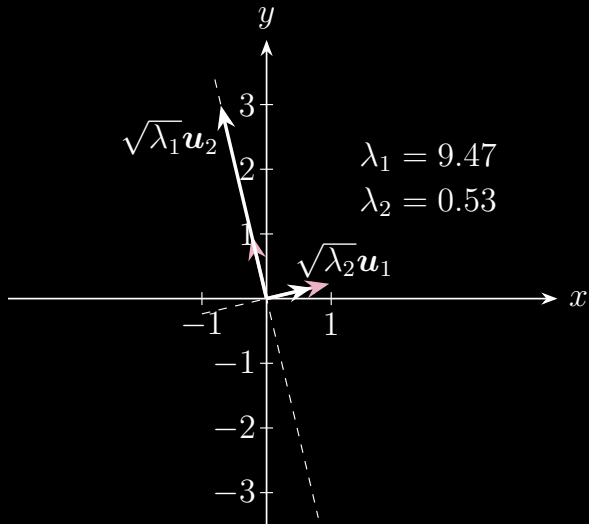
If $f(x, y) = \lambda_1 \lambda_2$, then the ellipse is

$$\lambda_1(\mathbf{u}_1^\top \mathbf{w})^2 + \lambda_2(\mathbf{u}_2^\top \mathbf{w})^2 = \lambda_1 \lambda_2 \quad \Rightarrow \quad \frac{(\mathbf{u}_1^\top \mathbf{w})^2}{(\sqrt{\lambda_2})^2} + \frac{(\mathbf{u}_2^\top \mathbf{w})^2}{(\sqrt{\lambda_1})^2} = 1$$

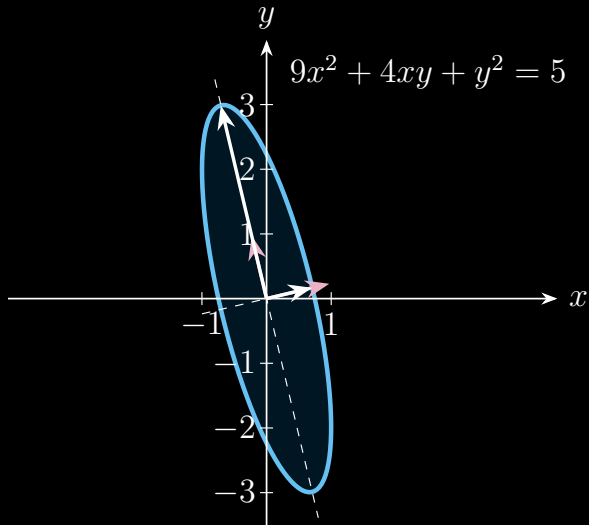
Ellipse $\frac{(\mathbf{u}_1^\top \mathbf{w})^2}{(\sqrt{\lambda_2})^2} + \frac{(\mathbf{u}_2^\top \mathbf{w})^2}{(\sqrt{\lambda_1})^2} = 1$



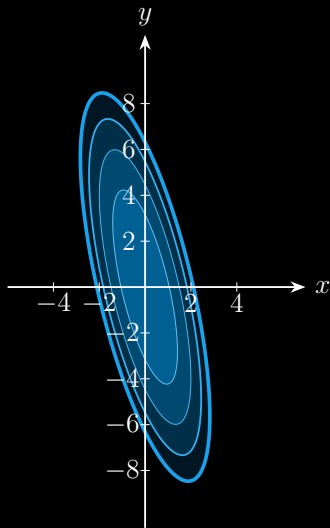
Ellipse $\frac{(\mathbf{u}_1^\top \mathbf{w})^2}{(\sqrt{\lambda_2})^2} + \frac{(\mathbf{u}_2^\top \mathbf{w})^2}{(\sqrt{\lambda_1})^2} = 1$



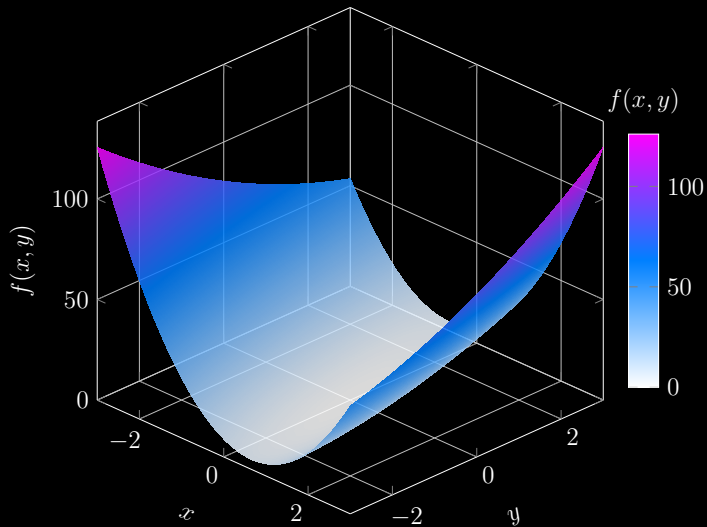
Ellipse $\frac{(\mathbf{u}_1^\top \mathbf{w})^2}{(\sqrt{\lambda_2})^2} + \frac{(\mathbf{u}_2^\top \mathbf{w})^2}{(\sqrt{\lambda_1})^2} = 1$



Quadratic Function $f(x, y) = 9x^2 + 4xy + y^2$



Quadratic Function $f(x, y) = 9x^2 + 4xy + y^2$



Thanks for your attention!

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